IE 303 MODELING AND METHODS IN OPTIMIZATION

The Computing Project

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Question 1)

a) When the problem is solved for positive integer variables the optimum solution is x* = (1 2) and the optimum value is $z^* = 35$.

b) LP relaxation of this problem is obtained by making the variables continuous. At first the LP has optimum solution x* = (1.28571, 0.857143) and $z^* = 27.85714286e{+}01$.

Then, the valid inequality is introduced to the model: $x_1 + x_2 \geq 3$

To obtain this valid inequality, Chvatal – Gomory Rounding Procedure was used. The two constraints from the original model ($4x_1 + x_2 \geq 6$ and $x_1 + 2x_2 \geq 3$) were added up and then divided by 3, and the floor values were taken. This procedure gave $x_1 + x_2 \geq 3$ valid inequality which automatically cuts the previous solution x* = (1.28571, 0.857143) because their summation is smaller than 3. However, it does not cut the optimum solution which is x* = (1 2).

Question 2)

a) This is a knapsack problem.

b) Let c = [1 2 3 4 5], a = [1 2 3 4 5], and b = 11. The optimal solution vector becomes x* = (1 1 1 0 1). The optimal value z* = 11.

c) When the given parameters c = [17 35 19 67 3 8 42], a = [5 11 8 18 21 24 18], b = 45 are added as an input to the code, the model becomes (with the cover inequalities):

$$\mathbf{max}\ \mathbf{17}x_1 + \mathbf{35}x_2 + \mathbf{19}x_3 + \mathbf{67}x_4 + \mathbf{3}x_5 + \mathbf{8}x_6 + \mathbf{42}x_7$$

Subject to $5x_1 + 11x_2 + 8x_3 + 18x_4 + 21x_5 + 24x_6 + 18x_7 \leq 45$

$x_7 \leq 0$

$x_6 \leq 0$

$x_5 \leq 0$

$x_1,\ x_2,\ x_3,\ x_4,\ x_5,\ x_6,\ x_7 \in [0,1]$

$The\ optimum\ vector\ becomes\ x* = [1, 1, 1, 1, 0, 0, 0]$ with the z value z*= 138.

Question 3)

Decision Variables:

$$x_{ij} = \begin{cases} 1, & if\ there\ is\ a\ path\ from\ agent\ i\ to\ agent\ j \\ 0, & otherwise \end{cases}$$

Objective Function:

$$\min \sum_{i \neq j} x_{ij}$$

Constraints:

$$\sum_{(\boldsymbol{i},\boldsymbol{j}) \in \boldsymbol{A}} \boldsymbol{x_{ij}} = \mathbf{1} \qquad \forall \boldsymbol{i}$$

$$\sum_{(\boldsymbol{k},\boldsymbol{i}) \in \boldsymbol{A}} \boldsymbol{x_{ki}} = \mathbf{1} \qquad \forall \boldsymbol{i}$$

$$\boldsymbol{x_{ij}} \in \{\mathbf{0}, \mathbf{1}\}, \qquad \forall (\boldsymbol{i}, \boldsymbol{j}) \in \boldsymbol{A}$$

The objective function minimizes the number of one-way channels used in a way that still guarantees all agents communicating with each other.

The first constraint shows that every agent can send a message and the second constraint shows that every agent can receive a message. The sending and receiving of a message can happen in a directed way or through other agents. However, in the final network, all of the agents have a way to connect with each other.

After putting this problem into the solver, the previous two-way network was:

Two-Way Channel Network (Undirected):

Agent 1 <-> Agent 2

Agent 1 <-> Agent 3

Agent 1 <-> Agent 4

Agent 1 <-> Agent 5

Agent 1 <-> Agent 6

Agent 1 <-> Agent 7

Agent 2 <-> Agent 3

Agent 2 <-> Agent 4

Agent 2 <-> Agent 5

Agent 2 <-> Agent 6

Agent 2 <-> Agent 7

Agent 3 <-> Agent 4

Agent 3 <-> Agent 5

Agent 3 <-> Agent 6

Agent 3 <-> Agent 7

Agent 4 <-> Agent 5

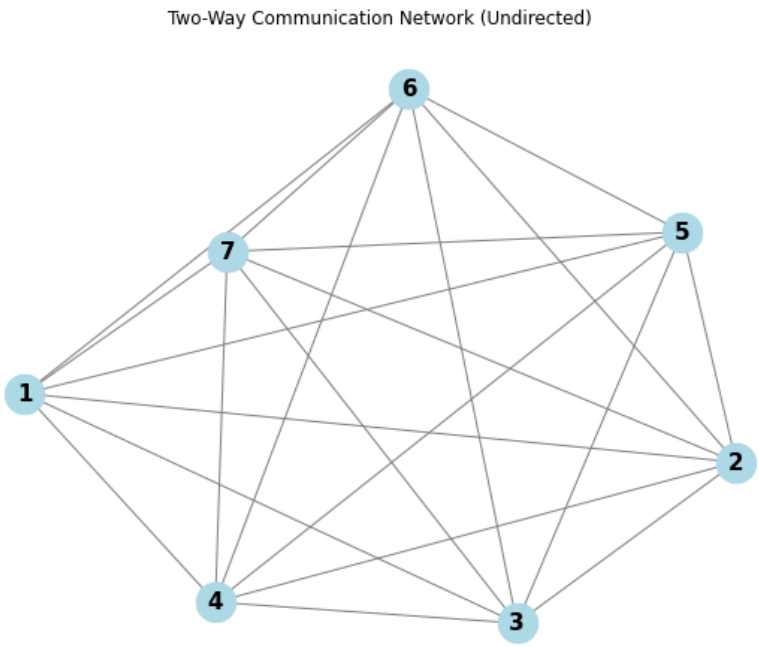
Agent 4 <-> Agent 6

Agent 4 <-> Agent 7

Agent 5 <-> Agent 6

Agent 5 <-> Agent 7

Agent 6 <-> Agent 7



The current one-way network is:

One-Way Channel Network (Directed):

Agent 1 -> Agent 5

Agent 2 -> Agent 3

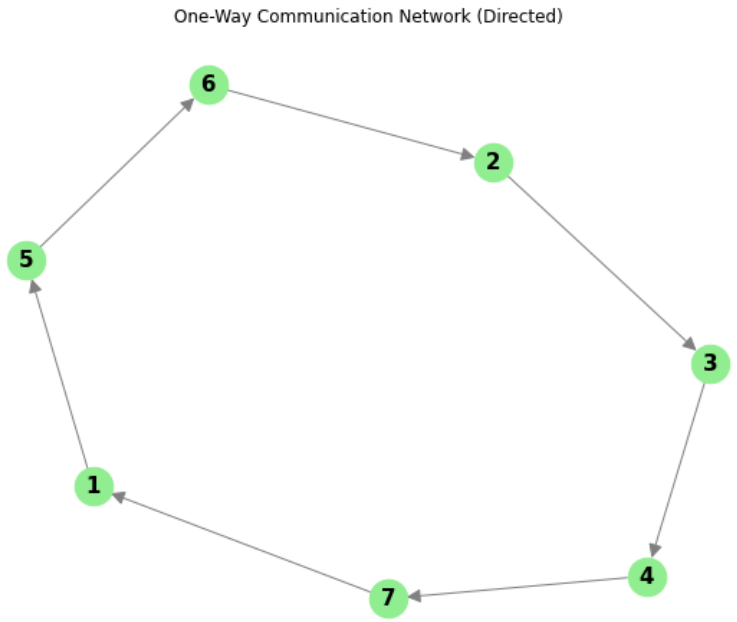
Agent 3 -> Agent 4

Agent 4 -> Agent 7

Agent 5 -> Agent 6

Agent 6 -> Agent 2

Agent 7 -> Agent 1



As seen in the One-Way Communication Network figure stated above, all the agents has a way of receiving and sending their secret messages.

Question 4)

a) **Parameters:**

n: the size of the magic square $(n \times n)$

M: the magic constant, the sum of the variables in every row, column, and diagonal

$$M = \frac{n(n^2+1)}{2}$$

Decision Variables:

x_{ij} = the value located in row i and column j

Objective Function: There is no objective function since we are not trying to maximize or minimize anything.

Constraints:

$$\sum_{j=1}^{n} x_{ij} = M, \qquad \forall i$$

(The sum of numbers in each row is equal to M.)

$$\sum_{i=1}^{n} x_{ij} = M, \qquad \forall j$$

(The sum of numbers in each column is equal to M.)

$$\sum_{i=1}^{n} x_{ii} = M, \qquad \forall j$$

(The sum of numbers in the main diagonal is equal to M.)

$$\sum_{i=1}^{n} x_{i,n-i+1} = M, \qquad \forall j$$

(The sum of numbers in the anti-diagonal is equal to M.)

$x_{ij} \neq x_{kl}$ for $(i,j) \neq (k,l)$

(No number is used twice.)

$x_{ij}, M \geq 0 \; \forall i,j$

b) 8×8 Magic Square

Parameters:

$n = 8$

$$M = \frac{8(8^2 + 1)}{2} = 260$$

Decision Variables:

x_{ij} = the value located in row i and column j where $x_{ij} \in \{1,2,3,\ldots,64\}$

Constraints:

$$\sum_{j=1}^{8} x_{ij} = 260, \qquad \forall i$$

$$\sum_{i=1}^{8} x_{ij} = 260, \qquad \forall j$$

$$\sum_{i=1}^{8} x_{ii} = 260, \qquad \forall j$$

$$\sum_{i=1}^{8} x_{i,8-i+1} = 260, \qquad \forall j$$

$x_{ij} \neq x_{kl}$ for $(i,j) \neq (k,l)$

$x_{ij} \in \{1,2,3,\ldots,64\}$

c) Magic Square (8×8) Found:

18	45	64	39	42	12	37	3
55	2	34	9	61	5	58	36
50	46	8	25	28	48	1	54
31	26	19	62	24	59	6	33
52	44	57	16	11	32	41	7
23	20	38	22	30	43	35	49
14	56	27	40	4	51	53	15
17	21	13	47	60	10	29	63

d) **Runtime for 3x3: 0.00 seconds**

Magic Square for 3x3:

2	7	6
9	5	1
4	3	8

Runtime for 4x4: 0.03 seconds

Magic Square for 4x4:

5	13	12	4
3	8	9	14
16	11	6	1
10	2	7	15

Runtime for 5x5: 1.21 seconds

Magic Square for 5x5:

18	9	13	10	15
1	16	6	17	25
21	22	3	14	5
2	7	24	20	12
23	11	19	4	8

Runtime for 6x6: 0.32 seconds

Magic Square for 6x6:

19	12	22	27	21	10
1	6	34	26	11	33
23	24	17	13	14	20
7	36	32	28	5	3
31	15	2	9	25	29
30	18	4	8	35	16

Runtime for 7x7: 8.52 seconds

Magic Square for 7x7:

32	5	46	42	14	9	27
26	15	38	45	12	8	31
41	19	30	6	47	7	25
28	36	13	18	23	22	35
24	16	10	29	20	37	39
3	44	34	2	48	43	1
21	40	4	33	11	49	17

Runtime for 8x8: 16.13 seconds

Magic Square for 8x8:

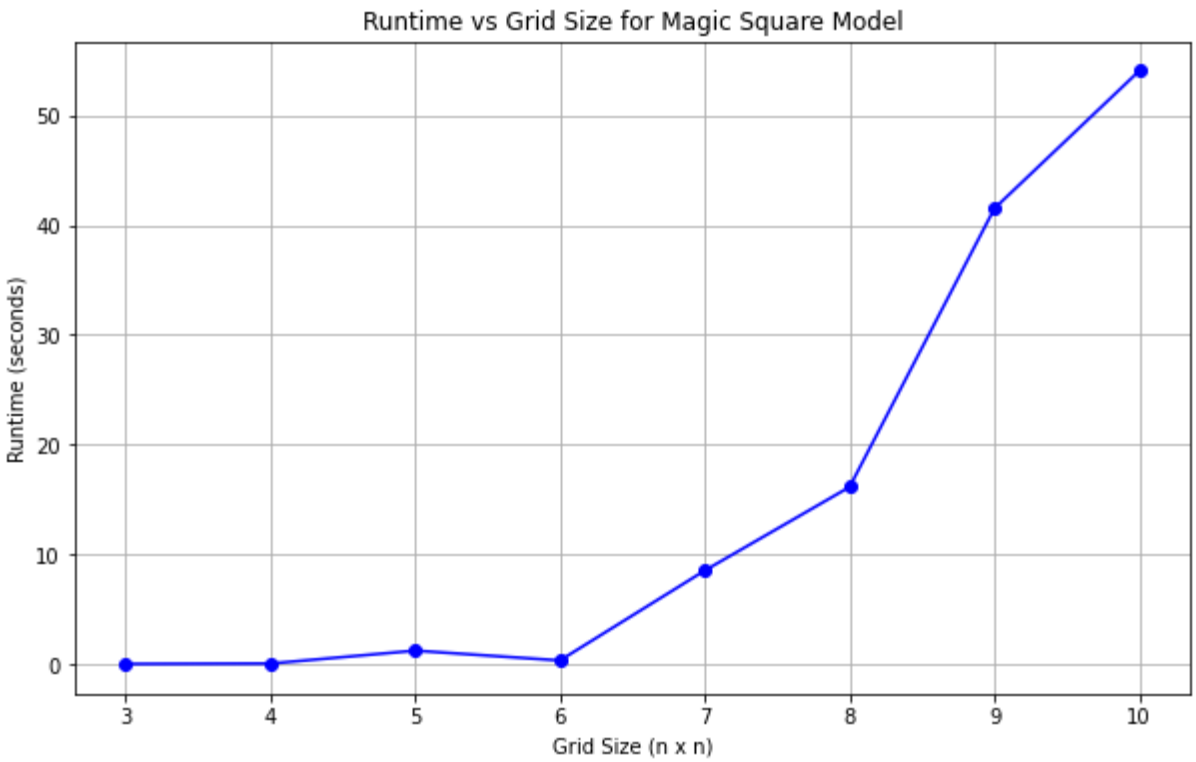
18	45	64	39	42	12	37	3
55	2	34	9	61	5	58	36
50	46	8	25	28	48	1	54
31	26	19	62	24	59	6	33
52	44	57	16	11	32	41	7
23	20	38	22	30	43	35	49
14	56	27	40	4	51	53	15
17	21	13	47	60	10	29	63

Runtime for 9x9: 41.53 seconds

Magic Square for 9x9:

61	27	77	11	25	66	16	78	8
74	9	19	10	72	57	40	55	33
52	28	60	1	3	65	68	43	49
36	45	50	80	71	24	17	41	5
23	47	32	64	39	30	76	4	54
44	62	31	53	26	7	56	48	42
6	35	21	75	46	38	67	18	63
51	79	59	2	58	13	14	12	81
22	37	20	73	29	69	15	70	34

Runtime for 10x10: 54.06 seconds

Magic Square for 10x10:

80	49	3	5	88	1	61	98	95	25
19	14	79	75	28	43	72	11	87	77
35	46	16	63	23	97	59	100	40	26
58	84	34	65	24	71	86	54	22	7
50	94	27	55	41	17	68	74	37	42
33	44	81	99	32	78	2	8	45	83
56	60	69	29	67	38	30	39	47	70
53	85	52	89	21	36	4	93	66	6
64	20	82	12	91	76	31	18	15	96
57	9	62	13	90	48	92	10	51	73